

Reviewer Report

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Title: Hybrid CNN-BiLSTM Attention Framework for Cross-Dataset DDoS Detection in IoT Networks with Explainable AI

The manuscript proposes a hybrid CNN–BiLSTM framework with an attention layer for DDoS detection in IoT networks, evaluated on four benchmark datasets (Apose-IoT-23, BoT-IoT, CIC-IoMT-2024 and CIC-IoT-2025) under binary and multiclass settings, and augmented with SHAP, LIME and ANOVA for explainability. The manuscript is well organised and the reporting is unusually detailed. However, the following points must be addressed.

- Figure 8(d), Figure 9(d), Figure 10(d) and Figure 12(d) all show that the highest-ranked input features for CIC-IoT-2025 are label1, label3, label4 and label_full. Section 6.5 states this explicitly: "The category label attributes (label4) ... are the primary features." These are the dataset's own annotation columns, so the network is being asked to predict the label from the label. This fully explains the reported 100.000 percent binary accuracy, the zero off-diagonal entries in Figure 5(d), and the fact that Figure 3(d) reports the best validation accuracy of 100.00 percent at epoch 1. All CIC-IoT-2025 results, including the multiclass 99.997 percent and every cross-dataset row involving that dataset, must be discarded and recomputed after the label columns are removed. The same audit must be applied to the other datasets. A complete list of retained and removed columns for each dataset should be published.
- Section 4.8 defines the partition as $D = D_{\text{train}} \cup D_{\text{test}}$ only, with a 75:25 ratio. Section 5.3 nevertheless states that "the best-performing model checkpoint was selected for evaluation based on the lowest validation loss", and Section 6.1 and Figures 3 and 4 report validation accuracy curves. Since no third partition exists, the validation set is the test set, and the reported figures are therefore obtained by selecting the checkpoint on the same data used for evaluation. A three-way split (train, validation, test) is required, with the checkpoint chosen on the validation partition and all reported metrics computed on an untouched test partition.
- Equations (3) and (39) construct one sequence per flow index using a look-back window of size T , so consecutive sequences share $T - 1$ of their T rows. Section 4.8 then applies a random 75:25 split and asserts that this "prevented data leakage". This is incorrect.
- Section 3.2 states that the mapping operator performs "zero-padding missing dimensions that are not present in dataset k ", which produces a union of the feature sets, whereas Equation (34) defines the unified space as the intersection over the four datasets. These two constructions are mutually exclusive. It is important to compare work with the following: PAACGAN: Protocol-Aware Generative Augmentation for Robust Imbalanced Network Intrusion Detection; TIDE-Net: A Two-Stage Temporal Deep Learning Framework for Multi-Granular IoT Intrusion Detection; Enhancing Intrusion Detection Systems with Synthetic

Attack Data and Advanced Classification Models; Enhanced network intrusion detection system for internet of things security using multimodal big data representation with transfer learning and game theory.

- From the supports in Table 6, a constant classifier that always predicts "Attack" achieves 99.987 percent accuracy on BoT-IoT and 97.939 percent on CIC-IoMT-2024. Every cross-dataset entry in Table 10 for these two targets (96.5, 97.2 and 97.4 percent for BoT-IoT; 95.9, 96.1 and 96.8 percent for CIC-IoMT-2024) is therefore worse than doing nothing, and the claim of "minimal degradation compared to the intra-dataset performance" is not supported. These discrepancies must be resolved.
- Table 5 reports the proposed framework on BoT-IoT as 99.99 percent accuracy, 99.99 percent F1-score and 99.99 percent recall. Recomputing from the confusion matrix in Figure 5(b) gives accuracy 99.998 percent but macro-F1 95.155 percent and macro recall 94.341 percent, exactly as printed in Table 4. The values in Table 5 correspond neither to the binary nor to the multiclass setting of Table 4 and appear to be the positive-class rather than the macro-averaged scores. Comparing a positive-class F1 against the macro F1-scores of other studies is not a valid comparison.
- Equations (16) to (18) compute a single energy vector from the fused representation, normalise it with a softmax over the D_{fused} dimensions, and multiply element-wise. There is no query, key or value, and no pairwise interaction between positions, so describing this as a "self-attention mechanism" in the abstract, in contribution (ii) and in Section 2.4 is not accurate; it is a feature-gating layer.
- All timing and energy figures in Tables 7 to 9 were obtained on a CUDA-enabled GPU, and Equation (51) defines energy as latency multiplied by an assumed 120 W, so the energy column carries no information beyond the latency column. The quoted per-sample latency of about 0.012 ms is the batch latency of 1.47 ms divided by the batch size of 128; it is an amortised throughput figure, not the per-flow detection latency a gateway would experience. No edge device was used, so the conclusion that the framework is "suitable for real-time edge-based IoT deployment" is not demonstrated.

Major revision is required.